Solution to Quiz 3.

[1]. (a) ① $\forall \varepsilon > 0$, $\lim_{n\to\infty} P(|X_n - X| > \varepsilon) = 0$

② $\lim_{n\to\infty} E[|X_n - X|^p]^{\frac{1}{p}} = 0$

(b) See. Assignment 6

(c) See Assignment 6

(d) A sequence $\{X_n\}_{n=1}^{\infty}$ is a Cauchy sequence in L^p, if

$\lim_{n\to\infty} \|X_n - X_m\|_p = 0 \quad (m > n)$

(e) Let $1 \le p < \infty$, if a family of random variables $\{X\}_{i\in I}$ satisf

$\lim_{M\to\infty} \left(\sup_{i\in I} \int_{\{|X_i| > M\}} |X_i|^p \, dP \right) = 0$

[2] (a) True, $\|\alpha X + \beta Y\| \le \alpha\|X\| + \beta\|Y\|$

(b) False; not for $0 < p < 1$. (c). True, $E[|X|^q] \le \left(E[(|X|^q)^{\frac{p}{q}}]\right)^{\frac{q}{p}} = \left(E[|X|^p]\right)^{\frac{q}{p}}$ (by Hölder inequality.)

(d). True

[3] (Weak law of large numbers) Let $\{X_n\}_{n=1}^{\infty}$ be i.i.d random variable such that. $E[X_i^2] < \infty$. Then

$$\frac{1}{n}\sum_{j=1}^{n} \xrightarrow{P} E[X_1].$$

proof: By Chebyshev's inequality,

$$P\left(\left|\frac{1}{n}\sum_{j=1} X_j - E[X_1]\right| > \varepsilon\right) \le \frac{1}{\varepsilon^2} \operatorname{Var}\left(\frac{1}{n}\sum_{j=1}^{n} X_j\right) = \frac{\operatorname{Var}[X_1]}{\varepsilon^2 n}$$

$\to 0$, as $n \to +\infty$.

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4. ① Since $Y_n \xrightarrow[n\to\infty]{P} Y$, there exists a subsequence $\{Y_{n_j}\}_{j=1}^{\infty}$ such that $Y_{n_j} \xrightarrow[j\to\infty]{a.s} Y$. By DCT, we have.

$$E[|Y|] = E[\lim_{j\to\infty} |Y_{n_j}|] = \lim_{j\to\infty} E[|Y_{n_j}|] \le E[|X|] < \infty$$

Hence, Y is integrable.

② Let $\{Y_{n_k}\}_{k=1}^{\infty}$ be an arbitrary subsequence of $\{Y_n\}_{n=1}^{\infty}$. Since $Y_{n_k} \xrightarrow[k\to\infty]{P} Y$, there exists a subsubsequence $\{X_{n_{k_j}}\}_{j=1}^{\infty}$ such that $Y_{n_{k_j}} \xrightarrow[j\to\infty]{a.s} Y$. Note that $|Y_{n_{k_j}} - Y| \le |Y| + |X|$, and $E[|X|+|Y|] < \infty$ by DCT. $\lim_{j\to\infty} E[|Y_{n_{k_j}} - Y|] = 0$. So we get $\lim_{j\to\infty} E[Y_{n_{k_j}}] = E[Y]$.

So for any subsequence, there exists subsubsequence with

$$E[Y_{n_{k_j}}] \xrightarrow{j\to\infty} E[Y],$$

we must have $\lim_{n\to\infty} E[Y_n] = E[Y]$.

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